

Mechanical Engineering Design Portfolio

BY Khushi JHA



Welcome! This portfolio showcases my journey as a mechanical design engineer, highlighting key projects that reflect my proficiency in CAD tools like CATIA and SolidWorks, and my passion for turning concepts into manufacturable solutions.



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Projects

Unicycle

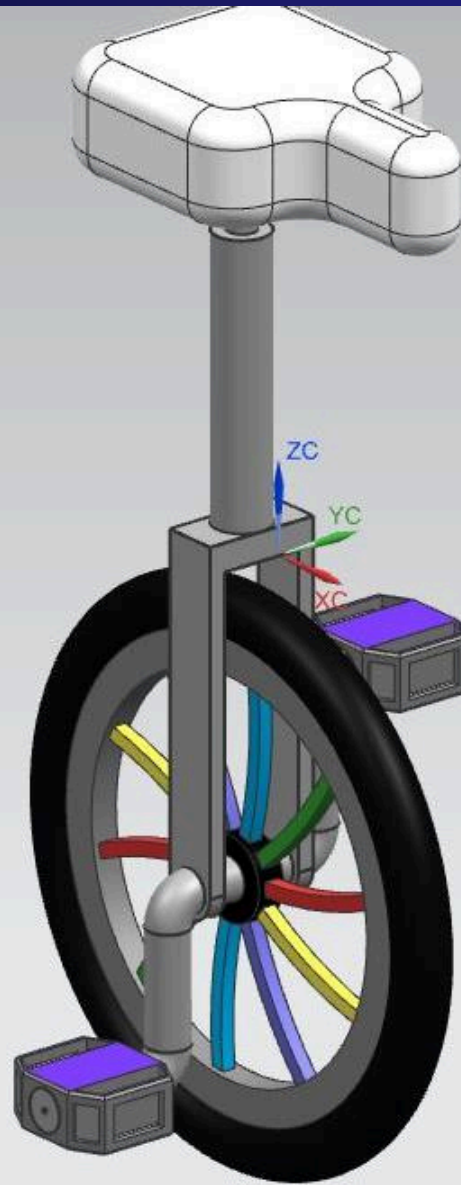
Epicyclic Gear Train

Bike Project

Globe Valve

Designed a functional unicycle assembly using CATIA, focusing on ergonomic structure and part integration.

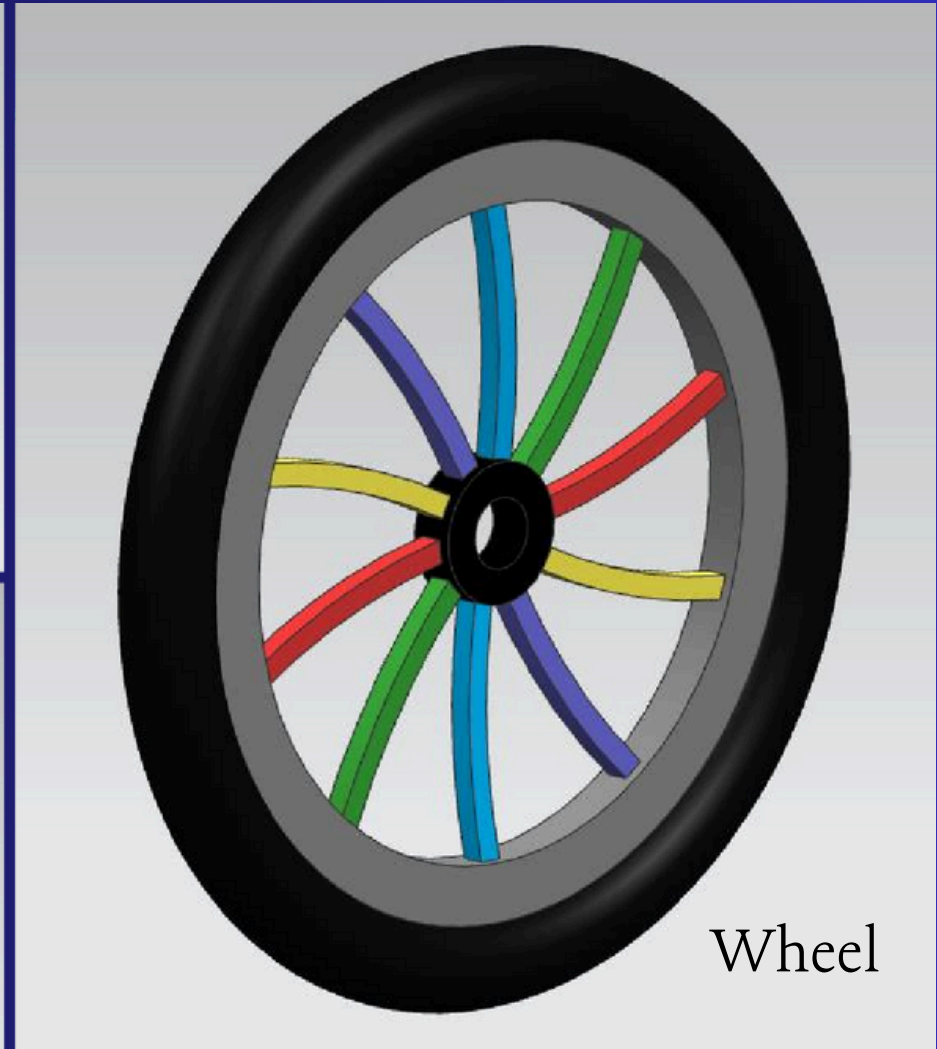
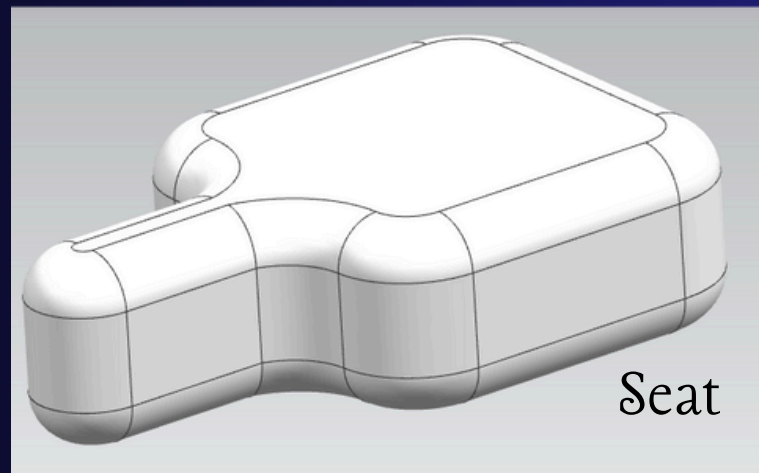
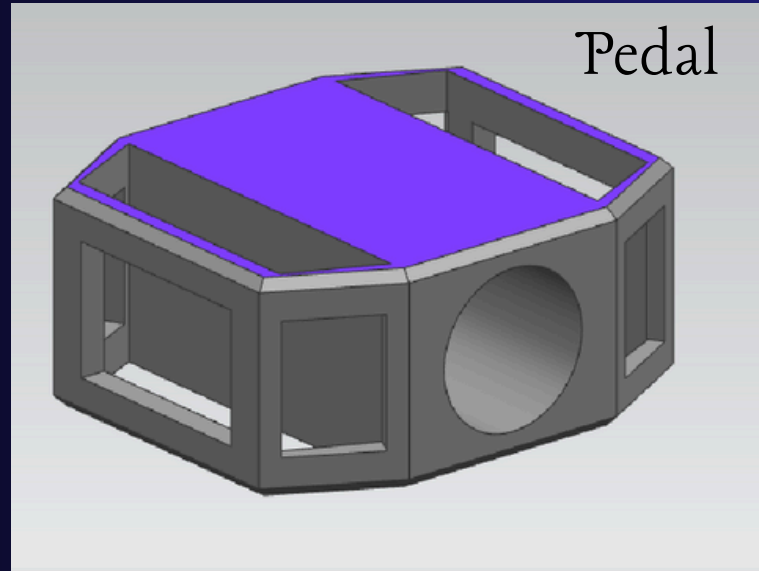
Unicycle Project



- ◆ Role: Mechanical Design Student
- ◆ Date: April 2024
- ◆ Tools Used: CATIA, SolidWorks
- ◆ Key Components: Axle, Pedal, Wheel, Seat, Frame

Unicycle Project

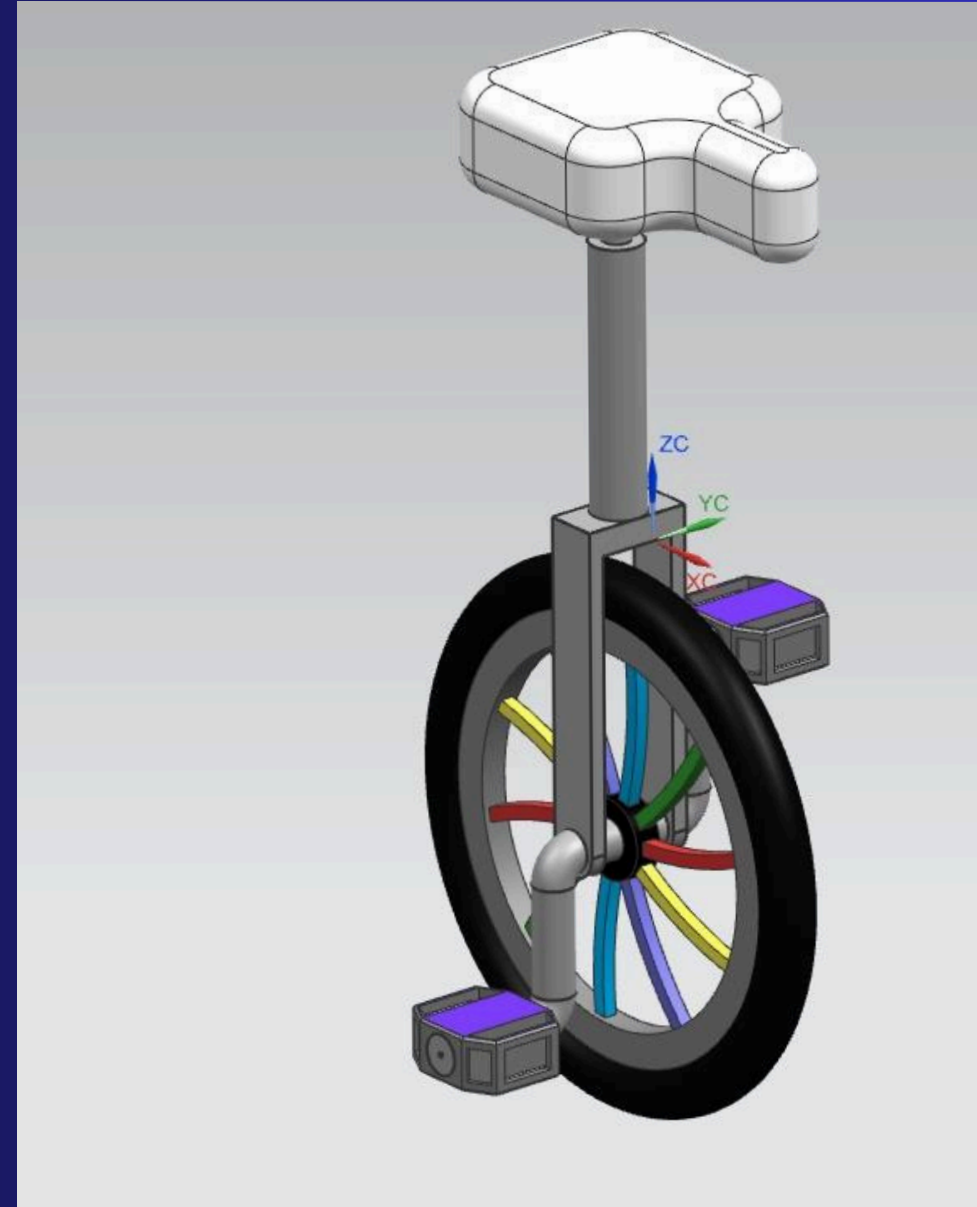
Part Files



Unicycle Project

• Full CAE Assembly with:

1. Axle
2. Pedal
3. Wheel
4. Seat



Bike Project

FullCAD Assemblywith:

- Front Wheel •
- Rear Wheel
- Frame
- Steering
- Pedals
- Seat
- Sprocket
- Chain



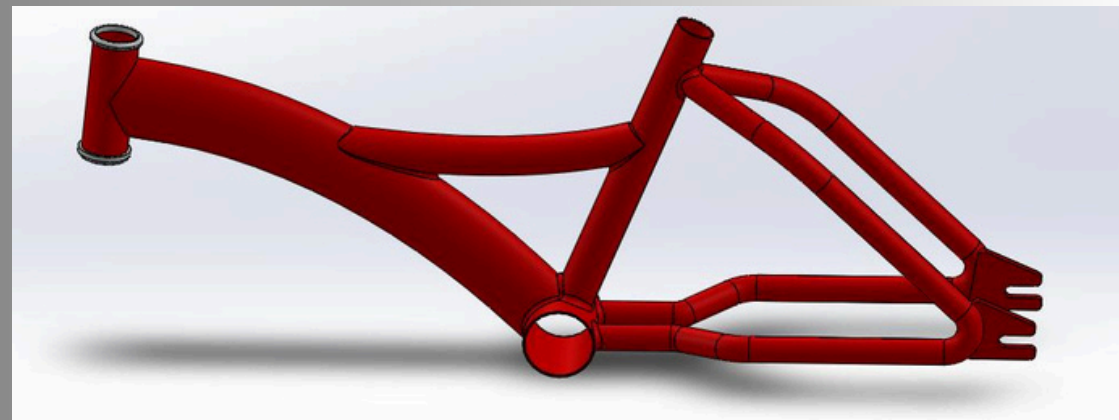
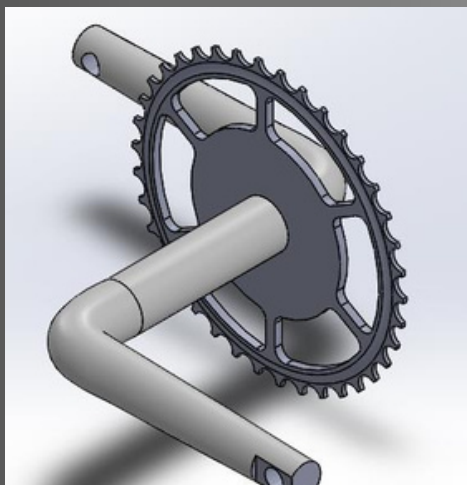


Bike Project

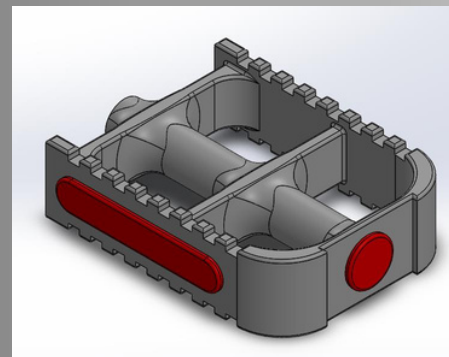
Steering Fork



Sprocket



Pedal



Seat



Rear Wheel



Front Wheel

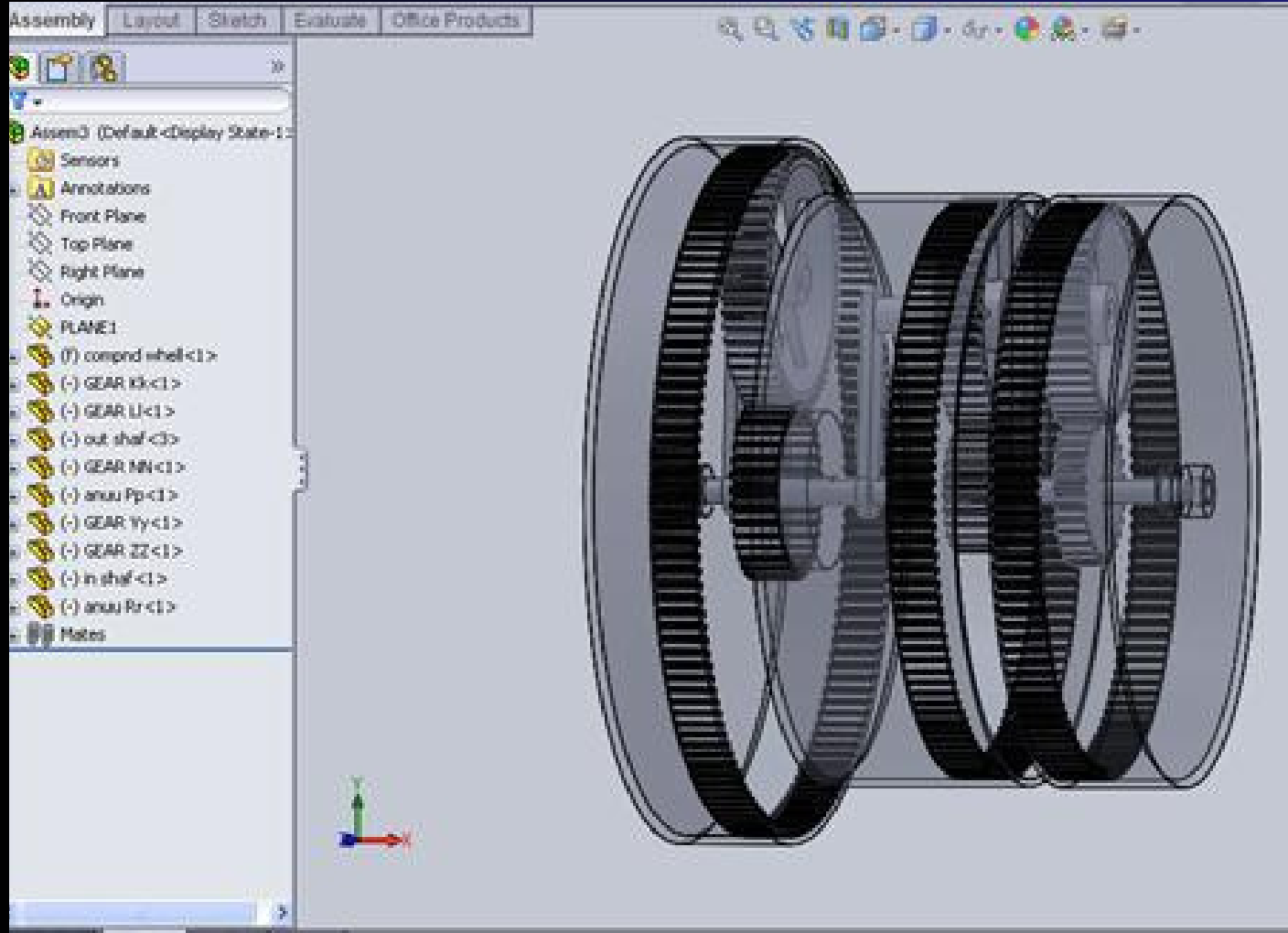


Chain

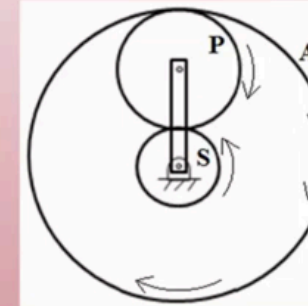


Design and Analysis of Epicyclic Gear Trains

Aim:- To Design Model and Analyze by Varying different Variables for a particular model for compound epicyclic gear train(with all its components)



Epicyclic Gear Train – Table Method



$$\frac{N_P}{N_S} = \frac{T_S}{T_P}$$

$$\Rightarrow N_P = \left(\frac{T_S}{T_P} \times N_S \right)$$

$$\Rightarrow N_P = \left(\frac{T_S}{T_P} \times 1 \right)$$

$$\Rightarrow N_P = \frac{T_S}{T_P}$$

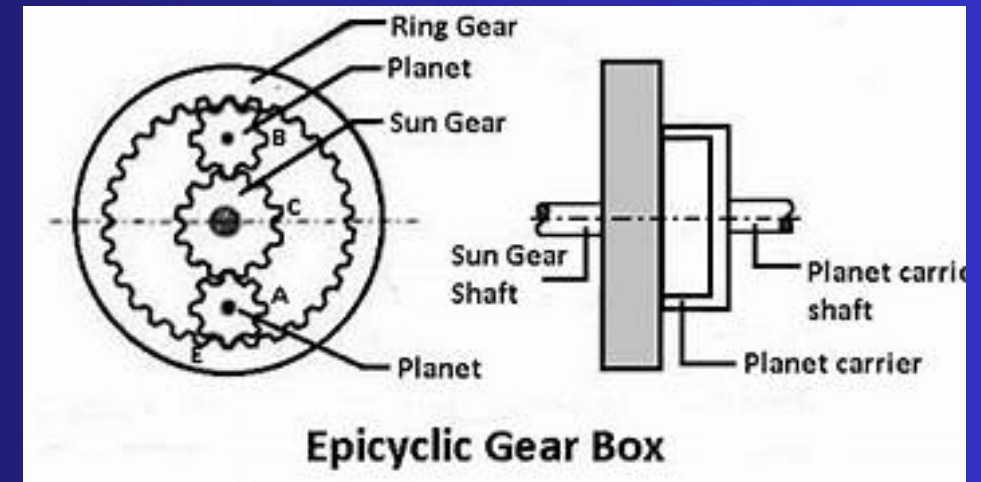
$$\frac{N_A}{N_P} = \frac{T_P}{T_A}$$

$$\Rightarrow N_A = \left(\frac{T_P}{T_A} \times N_P \right)$$

$$\Rightarrow N_P = \left(\frac{T_P}{T_A} \times \frac{-T_S}{T_P} \right)$$

$$\Rightarrow N_P = \frac{-T_S}{T_A}$$

Steps	Operation	Revolution of Elements			
		Arm	Sun Gear	Planet Gear	Annular Gear
1.	Fix the arm and give +1 rotation to the sun gear.	0	+1	$-\left(\frac{T_S}{T_P}\right)$	$-\left(\frac{T_S}{T_A}\right)$



Project Highlights: Compound Epicyclic Gear Train

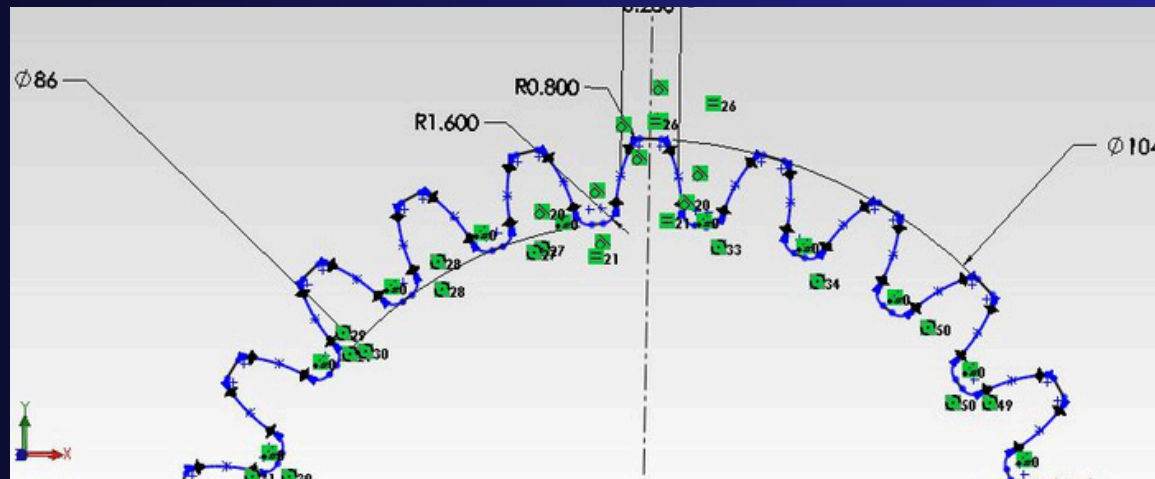
Modeled a compound epicyclic gear train in SolidWorks, incorporating realistic gear geometry and assembly constraints.

Defined gear parameters including number of teeth, pitch diameters, and annulus dimensions to achieve gear ratios between 3:1 and 6:1.

Conducted simulation-based analysis to evaluate gear interactions, stress distribution, and rotational dynamics under input torques ranging from 10 Nm to 50 Nm.

Performed analytical calculations to verify torque transmission, speed ratios, and power output, ensuring design feasibility.

Achieved 92% correlation between simulation and analytical results, validating the accuracy and reliability of the model.



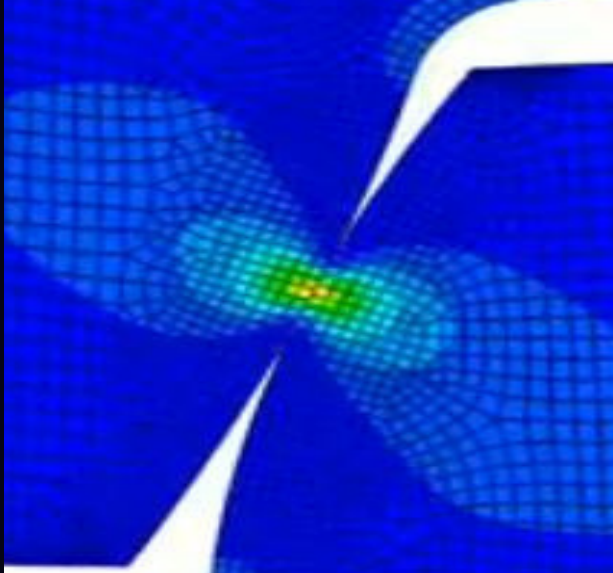
Analysis

By taking a particular material for the gear and by varying its module, power input etc, the different loads(tangential, dynamics, statics & wear tooth loads) were determined for all the geards(sun, planet & annulus)in the epicclic gear train.

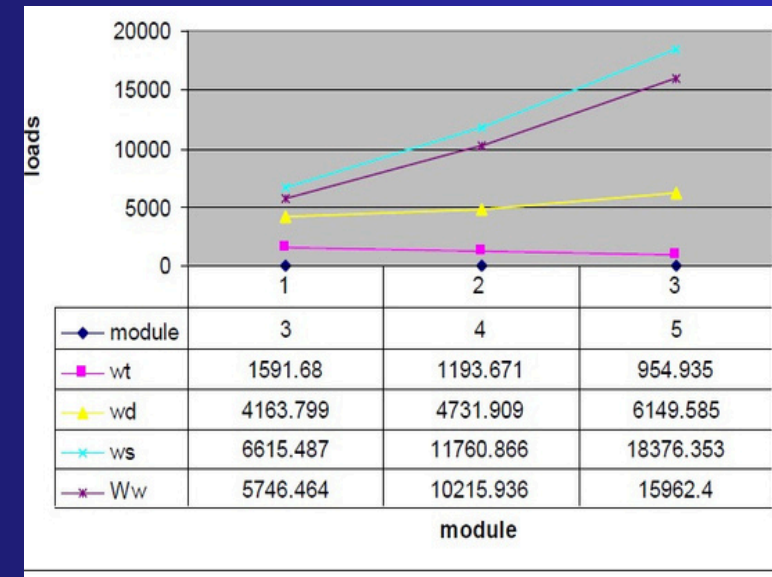
This was again done for various materials and all the different values were determined.

These values were them plotted into graphs and the results were analyzed.

A complete chart was created by which any gear with its preferable conditions could be selected by analyzing all the variables.



The above image shows the section of two gearstrain



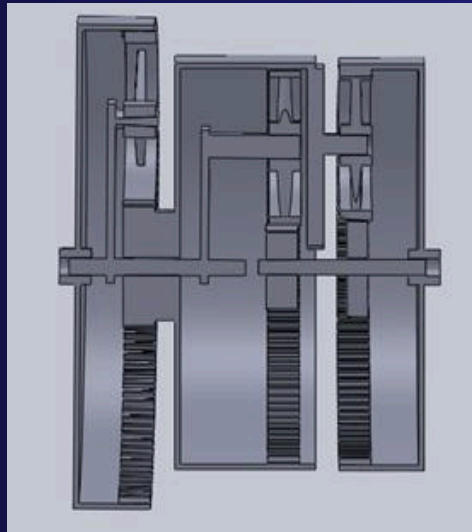
The image shows different loads acting in different modules of gears with particular power input and a particular material.

Results

The Complete Design with all the specifications of the gears were determined, all the loads were also calculated. A Complete working model was developed in solidworks CAD software with all its specifications . Analsis of the epicyclic Gear Train was successfull completed with all its results collected & presented in the form of charts.

The stresses were also calculated with the help of Ansys.

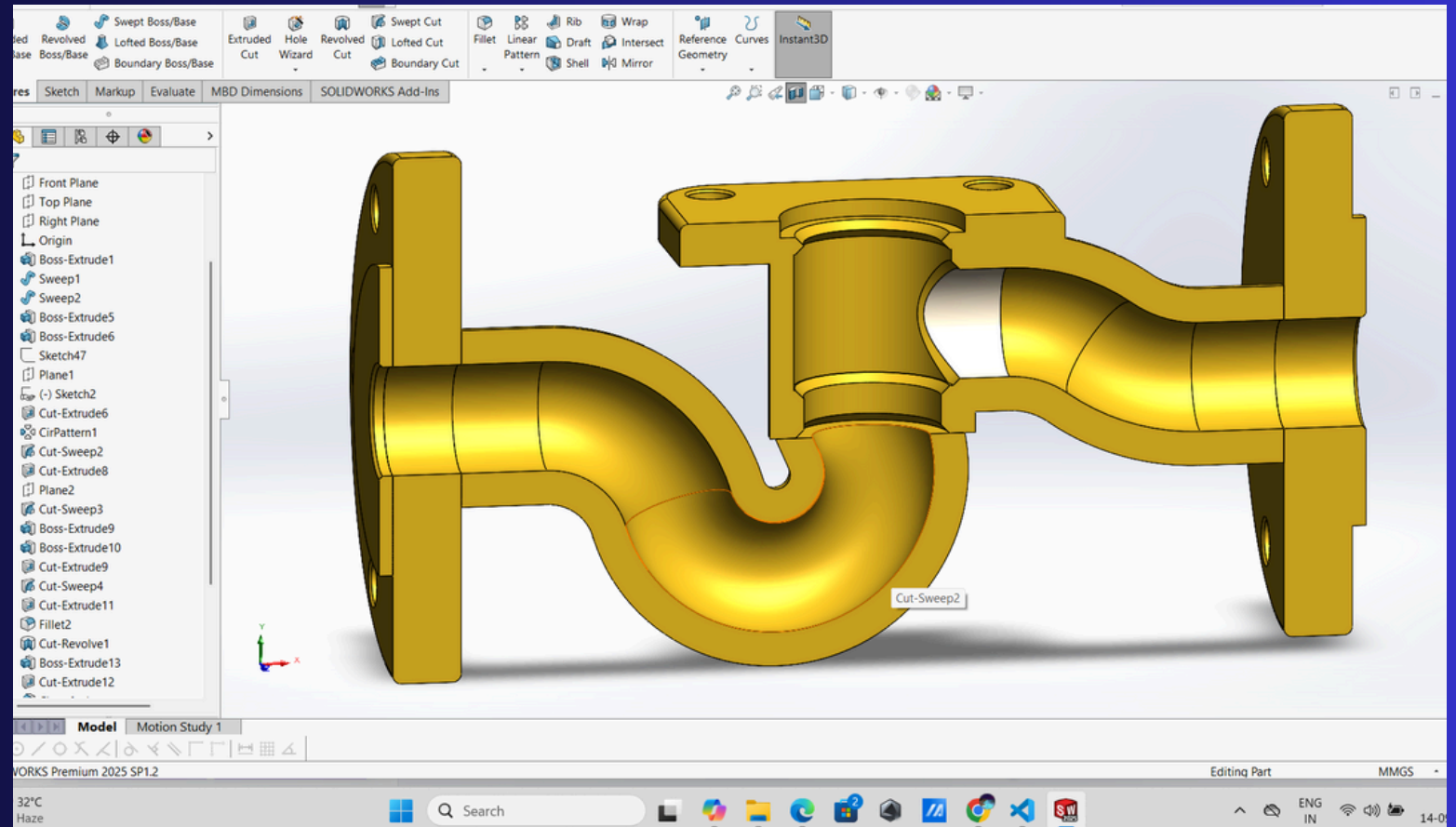
The loads with respect to the module, material and power could be seen from the graphs & a complete chart was created b which an gear with its preerrable conditions could be selected by analyzing all the variables affecting it.

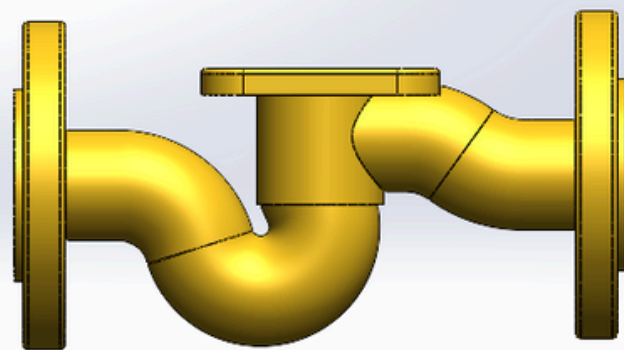


Globe Valve

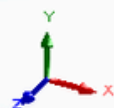
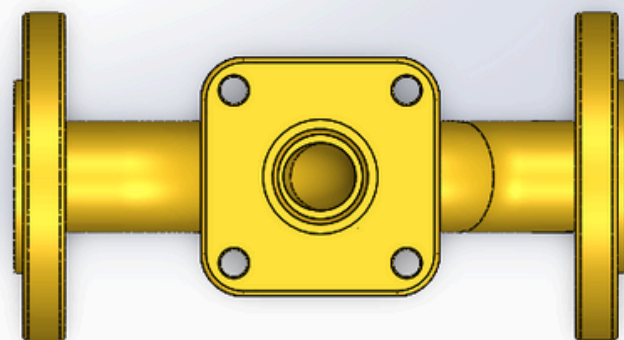
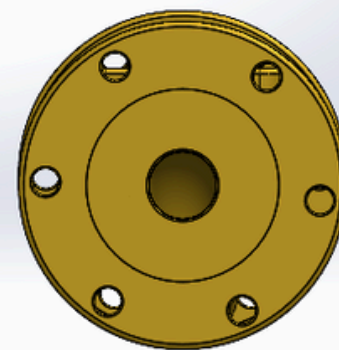
A globe valve is a linear motion valve used to regulate, start, or stop fluid flow in pipelines. Its design features a movable disk and a stationary seat, allowing precise control and effective throttling. Known for tight shut-off and minimal leakage, globe valves are widely used in steam systems, cooling circuits, and chemical processing where accurate flow modulation is essential.

◆ Tools Used: SolidWorks





*Left



*Trimetric

